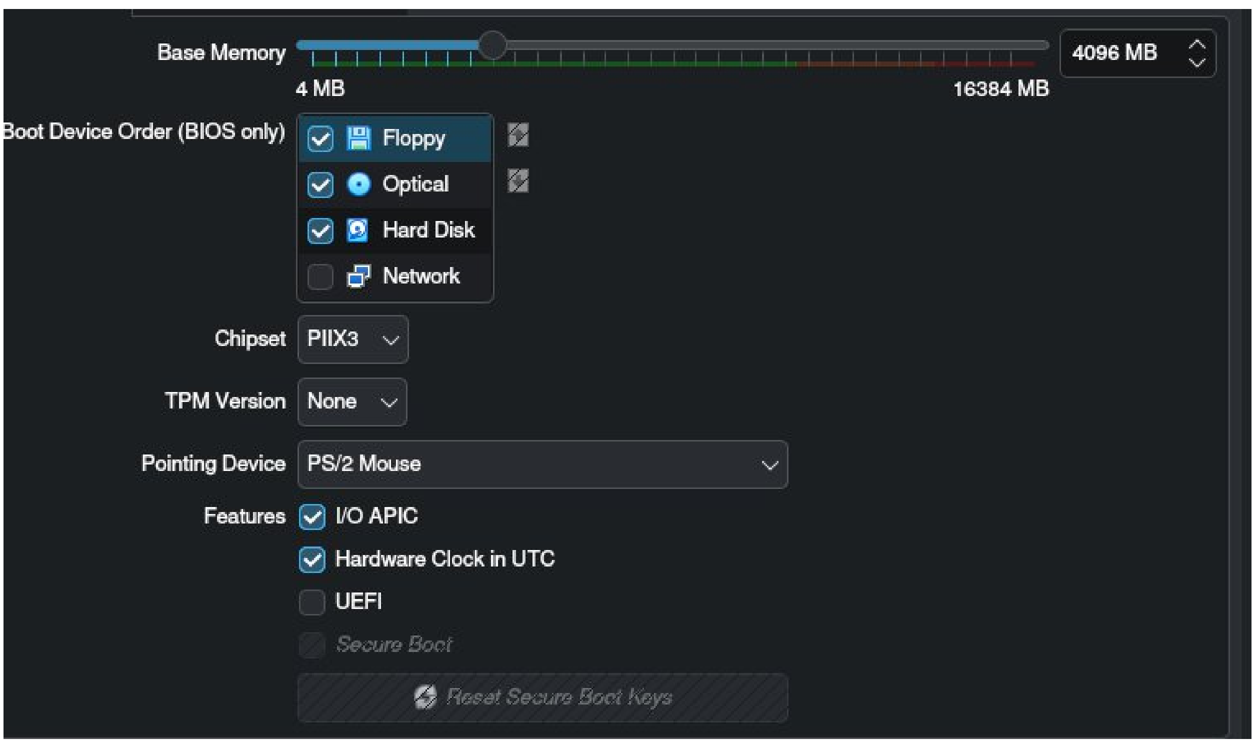
Wazuh SIEM

Setup and Configuration

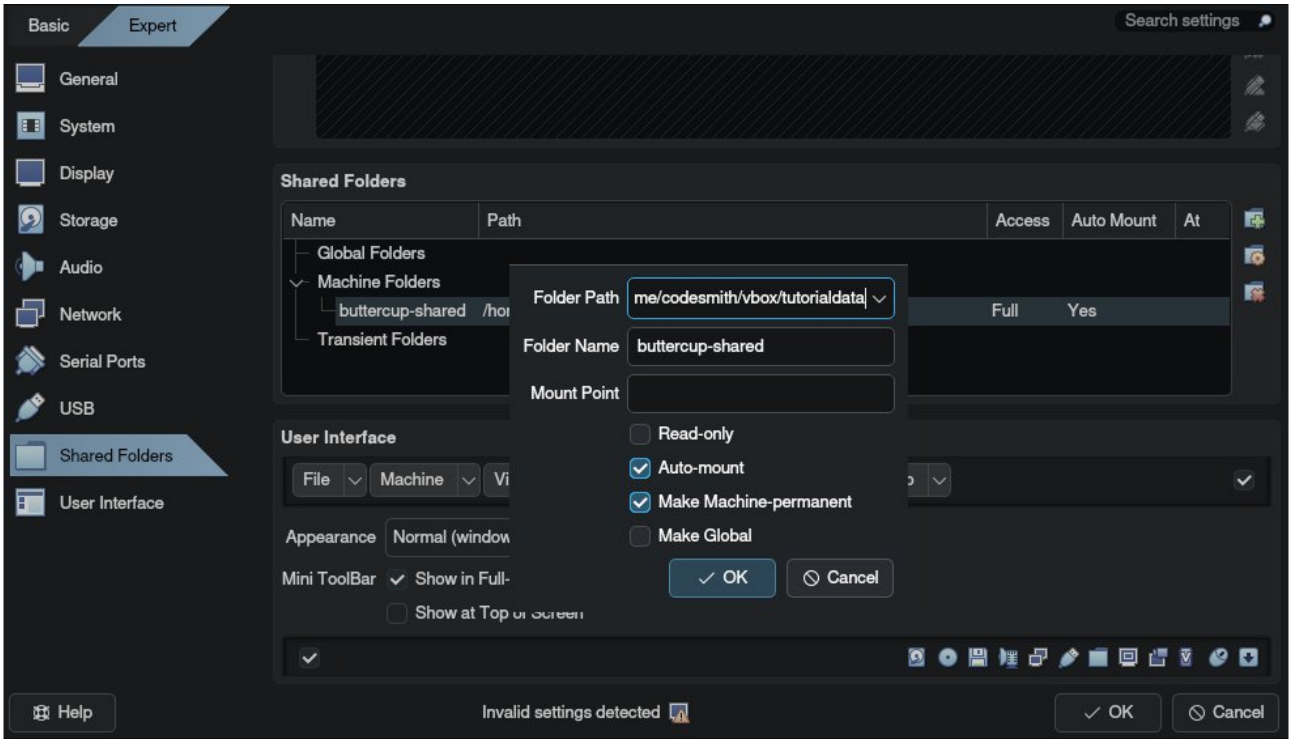
Project description
I was tasked with setting up a Security Information and Event Management (SIEM) tool to practice log analysis, threat detection.To complete this task, I performed the following steps to deploy the environment and ingest data:

Deploying and Configuring the Virtual Machine
I downloaded Oracle VirtualBox and imported a pre-built Wazuh .ova virtual machine file. Before starting the machine, I went into the VirtualBox System settings and adjusted the Base Memory to 4096 MB (4 GB). This was a critical step to ensure the VM had enough resources to run without errors.



Accessing the Wazuh Dashboard

I started the VM and configured a shared folder via the VirtualBox Devices menu. I linked my local tutorialdata folder and named the shared folder buttercup-shared, ensuring Auto-mount was checked.



After the VM booted, I logged into the command line using the root credentials and ran the necessary commands to fix the permissions for the shared folder.

```
usermod -aG vboxsf root

reboot
```

I then ran the `ifconfig` command to find the VM's IP address. Using this IP address, I navigated to the Wazuh web interface in my browser (https://the_local_ip_address) and logged into the dashboard using the default admin credentials.

Ingesting and Analyzing Data

To get the data into the SIEM for analysis, I went back to the VM's command line and navigated to the shared folder using:
cd /media/sf_buttercup-shared

I used the nano text editor to create a configuration file named ingest.yml.

```yaml
filebeat.inputs:

- type: log

  enabled: true

  paths:

    - /media/sf_buttercup-shared/mailsv/*.log

    - /media/sf_buttercup-shared/vendor_sales/*.csv

    - /media/sf_buttercup-shared/www1/*.log

    - /media/sf_buttercup-shared/www2/*.log

    - /media/sf_buttercup-shared/www3/*.log

output.logstash:

  hosts: ["localhost:5044"]
```

To process the logs and send them to the dashboard, I ran the following Filebeat command:

```
/usr/share/filebeat/bin/filebeat -c ingest.yml -e
```

Verifying the Data

Once the command finished, I returned to the Wazuh dashboard in my browser and navigated to Explore > Discover. I set the time range to Absolute with a start date of January 1, 2000, to ensure all historical data was captured. I typed * into the search bar and pressed Enter, confirming that the logs were successfully loaded and ready for querying.

Summary:

I successfully deployed and configured a local Wazuh SIEM environment using VirtualBox. I established a shared folder to securely transfer local files into the VM, created an ingestion configuration file, and used Filebeat to push the log data into the SIEM. Finally, I verified the data in the Wazuh dashboard, giving me a fully functional, locally controlled SIEM ready for security querying and threat investigation.